MIL-C-45096D(MU)
10 JANUARY 1967
SUPERSEDING
MIL-C-45096C (MU)
20 DECEMBER 1963

MILITARY SPECIFICATION

CARTRIDGE, 90MM, TP-T, M353A1 ASSEMBLING, MARKING AND PACKING

1. SCOPE

1.1 This specification covers the assembling, marking and packing for one type of cartridge designated as Cartridge, 90MM, TP-T, M353A1, for use in 90MM Guns, M36, M41 and M54.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 — Ammunition and Special Weapons, General Specification for.

MIL-I-45607 — Inspection Equipment, Supply and Maintenance of.

MIL-P-223 — Powder, Black.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).

MIL-STD-109 — Quality Assurance Terms and Definitions.

MIL-STD-414 — Sampling Procedures and Tables for Inspection by Variables for Percent Defective.

MIL-STD-1168 — Lot Numbering of Ammunition.

MIL-STD-1235 — Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ARMY

8861603 — Cartridge, TP-T, M353A1, for 90MM Guns, M36, M54 and M41.

FSC 1315

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7548467 — Container, Fiber, 90MM, M180A1, Assembly Details and Packing.

7548476 — Box, Packing, Ammunition for 90MM Gun Ammunition in Fiber Container, M180A1, Assembly, Details and Packing.

PUBLICATIONS

ARMY

ET-
8861603 — Equipment Tabulation.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Cartridges. The cartridges shall comply with all requirements specified on Drawing (dwg.) 8861603, and with all requirements specified in applicable specifications.

3.3 Propellant weight. The weight of the propellant shall not differ by more than plus or minus 0.25 ounce avoirdupois from that designated by the loading authorization for the particular lot of propellant used.

3.4 Assembly torque. The plug and disc assembly shall withstand subsequent reapplication of the minimum assembly torque specified on the applicable drawing without moving while thread sealing compound is still wet when tested as specified in 4.3.2.

3.5 Bullet pull. The cartridge case shall separate from the projectile at a load ranging from 5000 pounds minimum (min.) to 14,000 pounds maximum (max.) when tested as specified in 4.3.1.

3.6 Primers. The primer shall have the required number of flash holes as specified on the appropriate primer drawing.

3.7 Moisture content of black powder. The moisture content of the black powder at the loading station, at the time the igniter assembly is loaded into the cartridge case shall not exceed 0.60 percent when determined as specified in 4.3.4.

3.8 Proving ground. The cartridge shall comply with the following requirements when tested as specified in 4.3.3.

3.8.1 *Muzzle velocity.* The corrected muzzle velocity of the cartridge at ambient temperature shall not be more than 3075 feet per second (f.p.s.) nor less than 2925 f.p.s. and the standard deviation of the lot shall not be greater than 15 feet per second maximum.

3.8.2 *Tracer functioning.* The tracer shall ignite and there shall be visible evidence of the burning of the tracer composition to the 2000 meter target.

3.8.3 *Safety.* There shall be no hangfires or evidence of metal parts separation or breakup in the gunbore or in flight during any of the proving ground tests conducted.

3.9 Workmanship.

3.9.1 *Cartridge.* All manufacturing, processing, and assembly operations of the cartridge shall be correctly performed in a thoroughly workmanlike manner. All parts shall be free of cracks and burrs; and chips, dirt, grease, rust, or other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents.

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3.9.2 Care shall be exercised at all times to prevent the rotating bands from being nicked, burred or otherwise damaged.

3.9.3 The painting and marking shall be restored by the loading and assembling contractor on any component furnished to him painted and marked when such painting and markings have been marred or obliterated. (Bare spots in painting, other than marking, are allowable provided that they total not more than ¼ square inch in aggregate, and that the maximum dimension of any individual bare spot does not exceed 3/16 inch.)

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL–STD–109 in order to define the terms used herein. The provisions of Specification MIL–A–2550 shall apply.

4.1.1 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions specified within this specification.

(b) Number of units of product inspected.

(c) Results obtained, by defect code, for all inspections performed.

(d) Drawing, specification number and date, together with an identification and date of changes.

(e) Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.

(f) Number of items in the lot.

(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certification unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions.

4.2.1 *Lot formation.* The term "lot" as used throughout this specification refers to

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an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL–STD–105 and shall be numbered in accordance with MIL–STD–1168. Unless otherwise approved by the contracting officer, the inspection lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest monthly quantity contractually scheduled for delivery during the contract period. Inspection lots for components or subassemblies, other than the items of delivery, shall be homogeneous and of a size mutually convenient to both the contractor and the Government inspector. Each lot shall contain:

(a) Cartridge cases of one lot interfix number from one manufacturer.

(b) Projectile metal parts of one lot interfix number from one manufacturer.

(c) Propellant of one lot from one manufacturer.

(d) Loaded primers of one lot interfix number from one manufacturer.

(e) Loaded tracers of one lot interfix number from one manufacturer.

(f) Black powder of one lot from one manufacturer.

4.2.2 *Examination.* Inspection for Critical defects (and Major defect, when so specified), shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with MIL–STD–105 except that continuous sampling plans in accordance with MIL–STD–1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect, except where 100 percent inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Liner, prior to assembling into cartridge case (see dwg. 8880409 covering a detail of dwg. 8861603).

Categories	*Defects*	*Method of Inspection*	*Code No. (see 6.2)*
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 1.0 percent		
201.	Adhesive coverage incomplete in area indicated	Visual	03002
202.	Evidence of poor workmanship (see 3.9)	Visual	03005

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4.2.2.2 Cartridge case and liner assembly (see dwg. 8880409 covering a detail of dwg. 8861603).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 1.5 percent		
201.	Depth to liner from mouth of case	Gage	04001
202.	Gap between liner and case to extent that propellant can be entrapped	Visual	04002
203.	Improper surface of liner exposed	Visual	04004
204.	Evidence of poor workmanship (see 3.9)	Visual	04003

4.2.2.3 Primer, prior to assembly to cartridge case (see dwg. 8838153 covering a detail of dwg. 8861603).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:			
1.	Any flash hole missing (see 3.6)	Visual	01001
Major:	None defined.		
Minor:	None defined.		

4.2.2.4 Cartridge case with liner and primer, prior to loading propellant (see dwg. 8861603).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Primer fails to withstand disassembly torque (see 4.3.2)	Test	05001
102.	Exposed liner adhesive tacky or liner not secured to case in area of adhesive	Manual	05002
103.	Primer stake missing	Visual	05004
Minor:	AQL 0.65 percent		
201.	Foreign matter in interior of case (see 3.9)	Visual	05003

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4.2.2.5 Igniter assembly (see dwg. 7548695 covering a detail of dwg. 8861603).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Stitches missing or broken to extent that powder can escape	Visual	06001
Minor:	AQL 1.0 percent		
201.	Retaining tape missing	Visual	06003
202.	Evidence of poor workmanship (see 3.9)	Visual	06004

4.2.2.6 Support (see dwg. 8796851 covering a detail of dwg. 8861603).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 1.0 percent		
201.	Material thickness	Gage	07001
202.	Evidence of poor workmanship (see 3.9)	Visual	07002

4.2.2.7 Cartridge case with components prior to assembly to projectile (see dwg. 8861603).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Igniter assembly, disc or support missing or improperly assembled	Visual	08001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.9)	Visual	08002.

4.2.2.8 Projectile assembly prior to assembling cartridge case (see dwg. 8861603).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Sealing compound missing from threads of plug and disc assembly (prior to assembling tracer)	Visual	02006

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Categories	*Defects*	*Method of Inspection*	*Code No.*
102.	Plug and disc assembly fails to withstand minimum assembly torque (see 3.4 and 4.3.2)	Test	02001
103.	Tracer closing disc missing, damaged or loose	Manual	02002
104.	Adhesive missing or dry on cylindrical surface to rear of rotating band	Visual-Manual	02007
Minor:	AQL 1.0 percent		
201.	Marking incorrect or unidentifiable, including incorrect color code	Visual	02004
202.	Evidence of poor workmanship (see 3.9)	Visual	02005

4.2.2.9 Cartridge (see dwg. 8861603).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:			
1.	Primer above flush	Gage	09001
Major:	AQL 0.40 percent		
101.	Crimp missing or inadequate (to extent that projectile is liable to separate)	Visual	09013
Minor:	AQL 1.0 percent		
201.	Marking incorrect or unidentifiable including incorrect color code	Visual	09004
202.	Evidence of poor workmanship (see 3.9)	Visual	09006

4.2.2.10 Unsealed fiber container (see dwg. 7548467).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Packing component missing or out of place	Visual	10001
Minor:	AQL 1.0 percent		
201.	Glue or asphalt on cartridge	Visual	10002
202.	Cartridge cannot be removed by hand	Manual	10004

4.2.2.11 Sealed fiber container (see dwg. 7548467).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:	None defined.		

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Categories	*Defects*	*Method of inspection*	*Code No.*
Major:			
101.	Tape incomplete or badly wrinkled	Visual	11001
102.	Container cut or damaged through all impregnated layers	Visual-Manual	11002
103.	Metal end loose or distorted	Visual-Manual	11003
Minor:	AQL 2.5 percent		
201.	Tear tab length, min.	Gage	11004
202.	Color of tape incorrect	Visual	11005
203.	Marking incorrect or unidentifiable	Visual	11006
204.	Cuts, scuffs or gouges in outer layers	Visual	11007
205.	Gap between body and cover of container (exceeding ⅛ inch)	Visual-Manual	11008
206.	Contents loose in container	Manual	11009

4.2.2.12 Sealed wood packing box (see dwg. 7548476).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Box damaged exposing contents	Visual	12001
102.	DOD symbol or marking incorrect or unidentifiable	Visual	12002
103.	Board loose or broken	Visual-Manual	12003
104.	Hardware or strapping missing, broken or loose	Visual-Manual	12004
Minor:	AQL 1.0 percent		
201.	Car seal missing, unsealed or improperly attached	Visual	12005
202.	Handle missing or insecure	Visual-Manual	12007

4.2.3 *Testing.*

4.2.3.1 Check test for possible deterioration of primers. If the total elapsed time between the original acceptance of any primer lot and the assembly of that lot into complete round cartridges exceeds two years, or if the primers have been subjected to adverse conditions, however, brief, at any time since previous tests, the primer lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in the applicable specification immediately before the primer lot is assembled into the cartridges. The test shall be performed by the contractor responsible for assembling the cartridge (see 6.5).

4.2.3.2 Moisture content of black powder (see 3.7)—Major defect—Code No. 06005. The contractor shall provide adequate controls to assure the compliance of each lot with the applicable requirement. The contractor shall test black powder from at least

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one sample igniter assembly from each eight hours of complete round production. The black powder shall be tested as specified in 4.3.4 and any failure to comply with the applicable requirements shall be cause for rejection of the eight hours complete round production represented by the sample.

4.2.3.3 Propellant weight (see 3.3—Major defect—Code No. 05005. The propellant shall be weighed, identified and checkweighed 100 percent. The checkweighing shall be accomplished independently of the original weighing and, if performed manually, shall be performed by another operator. Any propelling charge which fails to comply with the applicable requirement shall be classed defective and removed from the lot. Equipment shall be in accordance with 4.2.4.

4.2.3.4 Profile and alignment (see dwg. 8861603)—Major defect—Code No. 09007. The cartridges shall be gaged 100 percent for profile and alignment. Any cartridge which fails to meet the applicable requirement shall be classed defective and removed from the lot. Equipment shall be in accordance with 4.2.4.

4.2.3.5 Bullet pull (see 3.5 and dwg. 8861603)—Major defect—Code No. 09008. The sampling plan and procedure for this test shall be in accordance with Standard MIL–STD–414, Section B, Part II, Table B–3, Code letter F, and an AQL of 1.5 percent. The sample taken shall be representative of each crimping machine used. The test shall be performed as specified in 4.3.1 with equipment in accordance with 4.2.4.

4.2.3.6 *Proving ground tests.* The following tests shall be performed as specified in 4.3.3.

4.2.3.6.1 Muzzle velocity (see 3.8.1)—Major defect—Code No. 09009.

4.2.3.6.1.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified, the sampling plan and procedure for this test shall be in accordance with Standard MIL–STD–414, Section B, Part II, Table B–3, Code letter I and AQL 1.0 percent. In addition, if the standard deviation of the sample cartridges multiplied by the factor 0.75 exceeds the lot requirement, the lot shall be rejected.

4.2.3.6.1.2 After three consecutive lots have met the criteria of 4.2.3.6.1.1, the sampling plan and procedure for this test shall be in accordance with Standard MIL–STD–414, Section B, Part II, Table B–3, Code letter F, and AQL 1.0 percent. In addition, if the standard deviation of the sample multiplied by the factor 0.65 exceeds the applicable lot requirement, the lot shall be rejected.

4.2.3.6.2 Tracer functioning (see 3.8.2)—Major defect, Code No. 09010.

4.2.3.6.2.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria of 4.2.3.6.1.1, the sample cartridges fired in the test for muzzle velocity shall be observed for tracer functioning. If three or more fail to comply with the applicable requirements, the lot shall be rejected.

4.2.3.6.2.2 After three consecutive lots have met the criteria of 4.2.3.6.1.1, the sample cartridges fired in the test for muzzle velocity (see 4.2.3.6.1.2) shall be observed for tracer functioning. If two or more sample cartridges fail to comply with the applicable requirements, the lot shall be rejected.

4.2.3.6.3 Target — impact dispersion (for information only) Code No. 09011. The cartridges fired for muzzle velocity in 4.2.3.6.1 shall be observed for target—impact dispersion.

4.2.3.6.4 Extreme temperature firing (for information only) Code No. 09012. A total

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of twenty cartridges shall be selected from each of the first three production lots and fired for muzzle velocity, target—impact dispersion, and maximum chamber pressure as specified in 4.3.3.2.

4.2.3.6.5 Safety (see 3.8.3)—Critical defect—Code No. 09013. All cartridges fired in the proving ground tests shall be observed for conformance with the applicable requirements. Failure of any sample cartridge shall be cause for rejection of the lot.

4.2.3.6.6 *Refiring.* If for any reason the proof director considers that the test conditions have detrimentally affected the test results he shall declare the results invalid and perform a new test on additional rounds as specified.

4.2.4 *Inspection equipment.* Equipment Tabulation Number ET–8861603 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.2.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed in the ET shall be in accordance with Specification MIL–I–45607.

4.2.4.3 *Government use of contractor's inspection equipment.* The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test methods and procedures.

4.3.1 *Bullet pull.* The security of the crimp of the cartridge case to the projectile shall be tested by applying a force in an axial direction at the rate of 0.100 to 0.125 inches per minute, inclusive, using test equipment in accordance with 4.2.4.

4.3.2 *Torque testing.* The torque shall be applied while holding the part immediately adjacent to the joint being tested to avoid possible loosening of other joints. The joint under test shall be marked to provide a means for detection of any movement between the adjacent parts. Any discernible movement between adjacent parts shall be regarded as a torque test failure.

4.3.3 *Proving ground.* These tests shall be performed at a Government proving ground in accordance with the applicable Acceptance Test Procedures. The cartridges shall be fired in a gun for which the round is standard and observation made for conformance to all applicable requirements (see 6.4).

4.3.3.1 *Velocity, tracer functioning, and target impact dispersion.* The cartridge shall be fired at a cloth target placed at a distance of 2000 meters from the muzzle of the gun. Muzzle velocity shall be determined and recorded for each round. Tracer functioning shall be observed for conformance to requirements and any failure shall be reported as either failure to ignite or approximate distance to burn out (estimated to nearest 500 meters). The hit pattern results in vertical and horizontal probable error in mils shall be recorded on the firing record for informational purposes (see 6.4).

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4.3.3.2 *Extreme temperature firing (for information only).* For this test ten rounds shall be temperature conditioned at —40° Fahrenheit and ten rounds shall be conditioned at 125° Fahrenheit for a period of 24 hours prior to firing. The cartridge shall be fired at a cloth target placed at a distance of 2000 meters from the muzzle of the gun. Muzzle velocity, maximum chamber pressure, and target impact dispersion data shall be recorded (see 6.4).

4.3.4 *Moisture content of black powder.* The moisture content of the black powder shall be determined in accordance with procedure specified in MIL–P–223.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A.* The cartridges shall be preserved and packaged in accordance with dwg. 7548467.

5.2 Packing.

5.2.1 *Level A.* The containers shall be packed in accordance with dwg. 7548476.

5.3 Marking. Marking of the packing container shall be in accordance with dwgs. 7548467 and 8796521. Marking of the packing box shall be in accordance with dwgs. 7548476 and 8796522.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

(a) Title, number and date of this specification.

(d) *Data cards.* Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL–STD–1167.

6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 *Inspection equipment lists.* Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 *Inspection equipment list codes.* The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD — Contractor design on controlled contractor format and/or commercial equipment.

MU — Army design, mandatory for use.

6.3.3 *Army designs.* Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and

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nonmandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with nonmandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL–D–1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 *Submission of contractor designs.* Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–NDI. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Submission of test data. In addition to the normal distribution of records, when the cartridge is procured by the Department of the Army, one copy of all ballistic test data shall be forwarded to each of the following:

Commanding Officer
Picatinny Arsenal
ATTN: SMUPA–ND2 and SMUPA–DC8
Dover, N.J.

6.5 Cost of check test. The contracting officer shall arrange for the contractor to be reimbursed for the expense incurred in the performance of the check tests. The test shall be conducted at Government expense without cost to the contractor who loaded the assemblies or to the contractor assembling the cartridge, and shall not constitute a basis for rejection against either contractor, except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer lot was under the jurisdiction of either contractor.

Custodian:

Army—MU

Preparing activity:

Army—MU

Project No. 1315-A545

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 OCT 64 1426

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PICATINNY ARSENAL
DOVER, NEW JERSEY 07801

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DEPARTMENT OF THE ARMY

COMMANDING OFFICER
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